TrashBot: Innovative Recycling By Utilizing Object Detection

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***Abstract*—With the ever-growing concern surrounding global warming, efficient waste management is a necessary practice to mitigate climate change. To address the problem, computer-vision-based systems are utilized with the integration of artificial intelligence (AI) and Internet of Things (Iot) technologies to enhance the waste segregation processes. To identify the waste, the object detection system is trained with a dataset consisting of 2401 images which are categorized into either paper, plastic or metal. This research employs the SSD MobileNet and EfficientDet D0 models. With various models trained, a comparative study is executed to examine each model's performance in classifying waste. These models are trained to recognize and classify waste into recycle categories - paper, plastic and metal, items failed to be classified are deemed as trash. The study aimed to incorporate hardware components to automate waste segregation, this is accomplished by designing an IoT hardware that incorporates the use of a servo motor. The servo motors allow the base of the bin to rotate depending on the classification of the object detected. This study was accomplished by loading the SSD MobileNet V2 onto the Raspberry Pi 4 Model B which achieved a precision of 0.714 and a recall of 0.809. With the SSD MobileNet V2 loaded, the python script allows the rotation of the bin. When the model detects metal the servo rotates 90 degrees, when paper is detected the servo motor rotates 180 degrees, when plastic is detected the servo motor rotates 270 degrees and when the item cannot be classified the servo motor will remain at 0 degrees**

Keywords— Artificial Intelligence Raspberry Pi, Object detection, Pre-trained models, Recycling

I. INTRODUCTION

The escalating threat of global warming necessitates urgent measures to combat climate change. One of the most effective methods to preserve the earth's environment is through recycling. By simply reusing material instead of discarding it, the need for natural resources decreases and the number of wastes is reduced.

Countries like Malaysia have been trying to increase the rate of recycling, which is quite low compared to its neighbouring countries. In 2023, Malaysia's neighbour Singapore, managed to achieve a recycling rate of 72% [1] whereas Malaysia only achieved a recycling rate of 35.38% [2] that very same year.

Over the years, the Malaysian Government have devised numerous plans to help increase the practice of recycling, from educating the youth to government-sponsored campaigns. With all this initiative it should be assumed that the citizens would be well versed in recycling. Therefore, since the public is probably educated in the manners of recycling, the reason why the recycling rate is low is due to cross-contamination.

Cross-contamination in the context of recycling refers to when materials are sorted into the wrong recycling bin, e.g. placing a glass bottle into the paper bin. This act is often due to aspirational recycling, whereby you simply throw something into the recycling hoping to find its way to where it needs to be [3]. When contamination occurs, recycling facilities will have to manually sort the garbage by hand and sometimes use filters to separate the objects. However, when contamination such as wet garbage mixing with recyclables, it can compromise the overall quality thus making it all non-recyclable. To prevent cross-contamination from occurring, proper waste management should be implemented as early in the chain. This is what the project aims to achieve.

This system will automatically identify and sort your waste. It will sort the waste into paper, metal, or plastic. Users will not have to overcomplicate themselves with the nuance of recycling.

II. LITERATURE REVIEW

There have been many researches conducted to classify and detect waste objects. These approaches often utilized sensor techniques, computer vision methodology or a mixture of both. Below are some descriptions of the aforementioned approaches.

Yang and Thung were one of the first researcher who implemented the use of computer-based vision to classify recyclable waste. These researchers studied the use of Support Vector Machine and Convolutional Neural Network to classify the image into six categories of garbage classes. From there they infer that SVM performed better than the CNN [4]

A system which implements the detection of small and medium-sized dry waste into the recyclable categories of glass, paper, metal, plastic, and non-recyclable (trash). This system utilises TensorFlow Lite, EfficientDet-Lite deployed on a Raspberry Pi 4 Model B, utilizing a Raspberry Pi camera version 1.3. With the Raspberry Pi connected to a laptop for programming, the camera captures images of waste objects placed inside a portable cardboard box. These images are processed in real time, enabling users to classify and identify waste objects effectively [5]

A smart garbage bin utilises a Convolutional Neural Network (CNN) to determine the waste category and sensors to prevent overloading the bin. An Arduino Nano board was used along with an ultrasonic sensor to observe the garbage levels in the bins. In the event that the garbage is full, the system sends out an SMS alert using a GSM module. A cloud-based system was also implemented to capture and store images to the cloud server for further training [6]

A study which focuses on optimizing waste classification using SSD-MobileNet, specifically targeting plastic bottles, glass bottles, and metal cans. The research involved training the model on a dataset of 952 images and deploying it on the Raspberry Pi 4 to evaluate the model's accuracy [7]

This paper aims to build upon these various techniques used for the detection and classification of waste through computer vision techniques and Internet of Things (IoT) technology to enhance the segregation process. The computer vision technique detects and classifies various types of waste, distinguishing between non-recyclable and recyclables into further categories of plastic, paper and metal. The hardware component ensures the waste is segregated into the allocated compartment to prevent cross-contamination.

III. METHODOLOGY

The system device consists of hardware and object detection models working seamlessly together. To establish a successful link between them, separate architectures are designed and integrated with the main goal of maximum efficiency.

A. Object Detection System

The dataset used in this work consists of 2401 images of JPEG, JPG and PNG extensions with 3 categories - paper, plastic and metal. The pictures were collected from Kaggle Repository [8], Google search engine and images were personally taken using an iPhone camera. Before the training phase, a partitioning strategy was employed, whereby 10% of the dataset was allocated for testing, while the remaining 90% was dedicated to training. Partitioning the dataset allows assessment of the performance of the model on unseen data. Figure 1 shows sample garbage images (top to bottom): metal, plastic and paper.

To train the system to recognize whether the object is metal, paper or plastic, it was necessary to label the images in the dataset. The labelling process was done through an image annotation tool, Labelimg. The purpose of the labelling was to teach the system how to recognize the object. The XML file stored the data of the labelled objects. The image size of the photo, location, coordinates, and the item name of the labelled object could be inspected in the file. The system read the XML file data and performed the training of learning procedures on the basis of the data. The steps, loss, and training time results were recorded. The trained files were then converted into the validation data files. Fig 2 illustrates the labelling process.

Fig. 1. Example of dataset

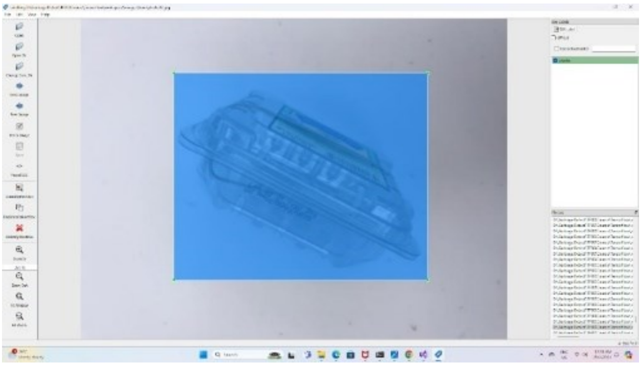


Fig. 2. Object labelling

TensorFlow 2 Detection Model Zoo provides a collection of detection models which were pre-trained on the COCO 2017 dataset. In this study, two pre-trained were selected from the zoo: SSD MobileNet V2 and EfficientDet D0. These models represent state-of-the-art object detection architecture commonly employed in computer vision tasks. The two models will be compared based on the accuracy, speed and performance on the Raspberry Pi 4 Model B

- SSD MobileNet v2 320x320
  MobileNetSSDv2 (MobileNet Single Shot Detector) is an object detection model with 267 layers and 15 million parameters. It provides real-time inference under compute constraints in devices like smartphones. Once trained, MobileNetSSDv2 can be stored with 63 MB,making it an ideal model to use on smaller devices. [9]

- EfficientDet-D0
  EfficientDet-D0 is a state-of-the-art object detection model for real-time object detection originally written in Tensorflow and Keras but now having implementations in PyTorch--this notebook uses the PyTorch implementation of EfficientDet. It has an EfficientNet backbone and a custom detection and classification network. Because of this backbone, EffcientDet is designed to efficiently scale from the smallest model size. The smallest EfficientDet, EfficientDet-D0 has 4 million weight parameters - it is truly tiny. EfficientDet infers in 30ms in this distribution and is considered and can be stored with only 17 megabytes of storage--making it both a small and fast model. [10]

To maintain consistency in the study, the same PC was utilized for training the models. The PC is equipped with an NVIDIA GeForce RTX 4060 8GB graphics processing unit (GPU) and runs on CUDA version 10.1. It operates on Python version 3.8.19 and TensorFlow version 2.13.0 for experimenting.

TABLE I. CHARACTERISTIC OF OBJECT DETECTION MODELS

Models	**Evaluation of Pre-trained Model**				
	Total loss	***Steps***	***Avg Precision (AP@IoU=0.50:0.95)***	***Avg Recall (AR@100)***	***F1-score***
SSD MobilenetNet v2	0.351	56000	0.714	0.809	0.758
EfficientDet D0	0.401	56000	0.496	0.712	0.586

The object detection system trained using SSD MobileNet V2 with 56,000 steps achieved a total loss value of 0.351. The model's average precision was 0.714, its average recall was 0.809, and it had an F1 score of 0.758. On the other hand, theEfficientDet D0 model, trained with 56,000 steps, resulted ina total loss value of 0.401. The average precision for EfficientDet D0 was 0.496, the average recall was 0.712, andthe F1 score was 0.586. Table 1 summarizes the evaluation performance of the two models.

To assess the model’s real-world applicability, the EfficientDet-D0 and the SSD MobileNet V2 was deployed onto the Raspberry Pi 4 Model B

B. Hardware Component

The proposed IoT solution requires several components to facilitate object detection and the segregation of different wastes. Table 2 lists the needed components and their function

TABLE II. LIST OF COMPONENTS USED AND THEIR FUNCTION

Component	***Function***
Raspberry Pi 4 Model B	Acts as the central processing unit, handling image processing, object detection and servo control
Raspberry Pi Camera Module 2	Captures video feed of the object to be sorted
MG996R Servo Motor	Rotates the base of the waste bin to align with the correct category segment. Rotates 90 degrees when metal is detected Rotates 180 degrees when paper is detected Rotates 270 degrees when plastic is detected Rotates 0 degrees when trash is detected
SG90 Servo Motor	Operates the drop hatch to release sorted waste into the correct bin segment

The system begins with the Raspberry Pi Camera Module 2, which captures a live video feed of the objects that need to be sorted. This camera, connected to the Raspberry Pi 4 Model B via the CSI port, provides video input for processing. The captured video is sent to the Raspberry Pi 4 Model B, which acts as the central processing unit. It handles image processing, object detection, classification, and servo motor control.

Once the video feed is received, the Raspberry Pi utilizes a pre-trained object detection model to analyze the frames. Each frame is processed to detect and classify objects into predefined categories: metal, paper or plastic. The detection model assigns a probability score to each classification. If the probability score is 0.7 or higher, the object is classified into one of the specified categories. Objects with a score below 0.7 are classified as trash.

Based on the classification results, the Raspberry Pi sends signals to the MG996R servo motor to rotate the base of the waste bin to the appropriate segment. The waste bin is divided into four segments, each corresponding to a category: trash (0 degrees), metal (90 degrees), paper (180 degrees), and plastic (270 degrees). The servo motor rotates the bin so that the segment aligns with the opening and corresponds to thedetected object’s category. Once aligned, the SG90 servo motor rotates thus dropping the object into the correct segment. This process repeats for each detected object, ensuring efficient and automated waste sorting. Figure 3 illustrates the sketch of the proposed hardware system and Figure 4 illustrates the flowchart of the IoT hardware system

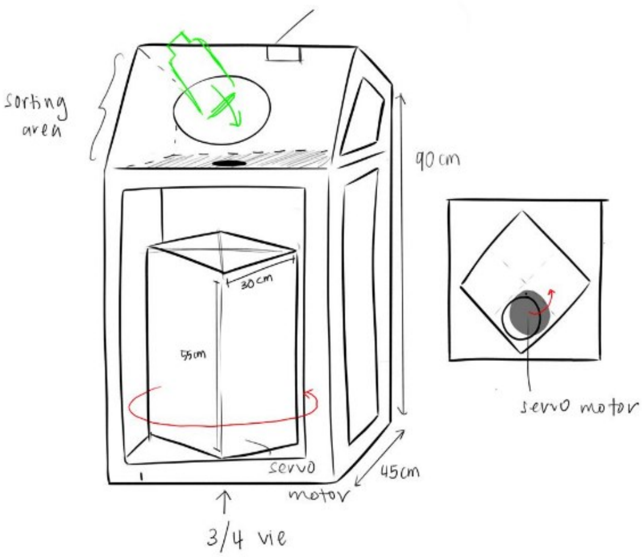


Fig. 3. Sketch of hardware system

Fig. 4. Figure of flowchart of IoT hardware system

IV. RESULTS AND DISCUSSION

Between the two models, only the SSD MobileNet V2 was the only one that was able to properly function on the Raspberry Pi 4 Model B. The EfficientDet-D0 model was unable to detect the garbage as it was likely due to the model being too computationally heavy for the Raspberry Pi 4 Model B. During the testing, the camera feed would often lag significantly, resulting in a noticeable delay between real-time and their detection screen. The lag rendered the EfficientDet-D0 ineffective for real-time application.

A canvas was attached to the servo motor to demonstrate the functionality of the object detection and the IoT hardware. The canvas was divided into four quadrants: red, blue, yellow and white. When no object is detected the canvas points to the white quadrant which remains at the origin point. However, when the servo motor rotates to 90 degrees, the canvas points to the red quadrant (metal), a 180-degree rotation points to the blue quadrant (paper) and a 270-degree rotation points to the yellow quadrant (plastic). Figure 4, 5 and 6 illustrates the detection of metal, paper and plastic respectively.

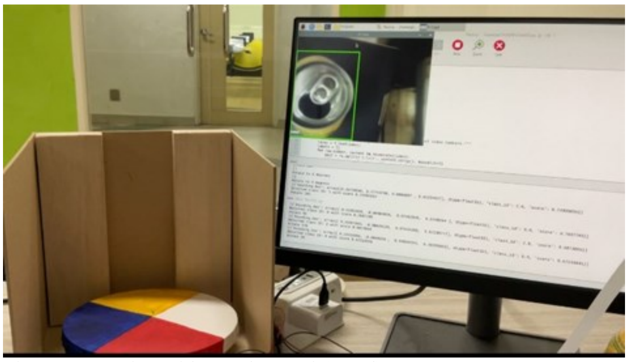


Fig. 5. Detection with Raspberry Pi (metal)

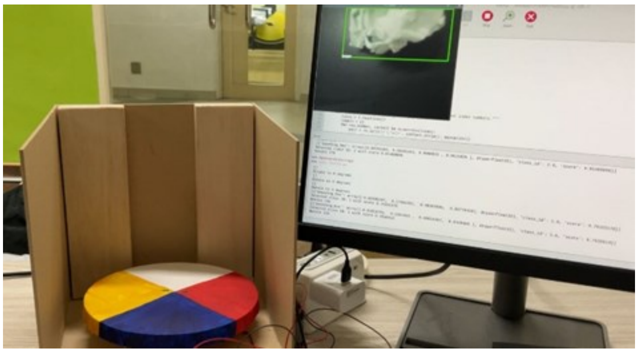


Fig. 6. Detection with Raspberry Pi (paper)

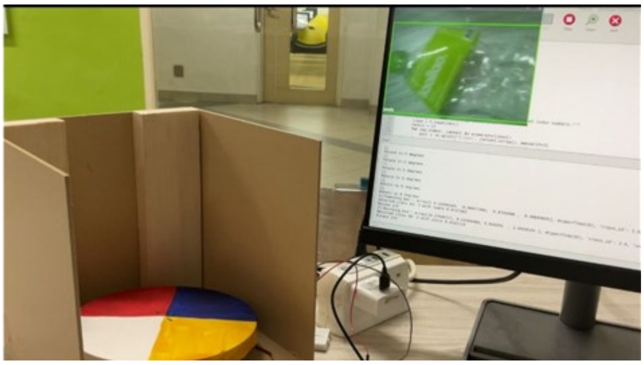
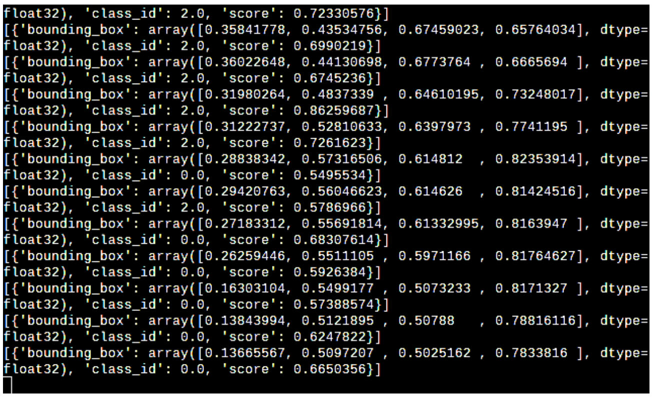


Fig. 7. Detection with Raspberry Pi (plastic)

The MG996R motor, which controls the base of the bin managed to rotate based on the classification of waste. Whereas the SR90 could not be implemented as planned due to the power supply limitation of the Raspberry Pi 4 Model B.

When deployed on the Raspberry Pi, the SSD MobileNet V2 model provided real-time classification results through the script Python shell. When garbage is detected, the python script will produce a message containing the class ID, score and the (x,y) coordinates of the bounding box. This message is produced roughly every 4 seconds when the bounding box detects the object. Figure 8 illustrates the messages produced in the shell of the Python script when garbage is detected



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float32), 'class_id': 2.0, 'score': 0.72330576}]
[{'bounding_box': array([0.35841778, 0.43534756, 0.67459023, 0.65764034], dtype=
float32), 'class_id': 2.0, 'score': 0.6990219}]
[{'bounding_box': array([0.36022648, 0.44130698, 0.6773764 , 0.6665694 ], dtype=
float32), 'class_id': 2.0, 'score': 0.6745236}]
[{'bounding_box': array([0.31980264, 0.4837339 , 0.64610195, 0.73248017], dtype=
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[{'bounding_box': array([0.31222737, 0.52810633, 0.6397973 , 0.7741195 ], dtype=
float32), 'class_id': 2.0, 'score': 0.7261623}]
[{'bounding_box': array([0.28838342, 0.57316506, 0.614812  , 0.82353914], dtype=
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[{'bounding_box': array([0.29420763, 0.56046623, 0.614626  , 0.81424516], dtype=
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[{'bounding_box': array([0.13665567, 0.5097207 , 0.5025162 , 0.7833816 ], dtype=
float32), 'class_id': 0.0, 'score': 0.6650356}]
```

Fig. 8. Message produced in shell

To test the overall accuracy of the model in real time deployment, a test was conducted by using specific object from each category: a metal can for the metal category, a crumpled piece of paper for the paper category and a plastic water bottle for the plastic category. During these tests, the model was set to process and generate 100 classification messages for each object.

For each category, we recorded the number of times the model correctly identified the object. For instance, when testing with the crumpled paper, the model correctly classified it as paper 72 out of 100 times, resulting in an accuracy of 72% for the paper category. This procedure was repeated for the metal and plastic categories, and the results were tabulated in Table III.

Additionally, an interesting observation was made regarding the detection behaviour of the model during the live object detection. Without a proper source of light, the model failed to produce a detection box around the garbage. Figure 9 illustrates the lack of a detection box around the object

TABLE III. ACCURACY OF SSD MOBILENET DEPLOYED ON RASPBERRY PI

Class	Total Message	Correctly classified	Accuracy
Paper	100	72	72%
Plastic	100	100	100%
Metal	100	16	16%

Fig. 9. Lack of Detection Box

V. CONCLUSION

The research aimed to investigate the performance of pre-trained models, SSD MobileNet V2 and the EfficientDet-D0and the integration of IoT hardware which manually segregates garbage. This research managed to integrate the SSD MobileNet V2 with the IoT hardware. The SSD MobileNet V2 was the most reliable as it achieved an averageprecision of 0.714 and an average recall of 0.809. The IoT managed to rotate the base of the bin with the MG996R servomotor based on garbage detection.

Moving forward, future research should attempt to train and test other object detection models. Specifically, models suchas YOLOv5, YOLOv8 and YOLOv9. By training other detection models a comparison can be made to determine which model is superior in detecting and classifying plastic, paper and metal.

Further improvements can be made to the hardware component. Due to the power limitation of the Raspberry Pi 4 Model B, the SR90 would require an external power source and for better accuracy, the system would benefit from a light source. Additionally, the accuracy could possibly be improved if a better camera was used for the object detection

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